1.

```
Create a report that displays EMPNO,ENAME,DEPTNO of employees who work with
that Employee.(Nested Sub Query)
```

```
SQL> select eid,first_name,dept_id
  2  from emp
  3  where first_name = (select first_name from emp where first_name='David');

       EID FIRST_NAME              DEPT_ID
---------- -------------------- ----------
       106 David                        80

SQL>
```

```
Display the employees who earn more than the avg salary of EMP table.( Nested
Sub Query).
```

2.

```
SQL> select eid,first_name
  2  from emp
  3  where sal > ALL(select avg(sal) from emp);

       EID FIRST_NAME
---------- --------------------
       100 Steven
       101 Neena
```

```
Display the ENAME,JOB  who are  Managers( Use Exists Operator)
```

3.

```
SQL> select first_name,job_id
  2  from emp
  3  where   EXISTS (select mgr from emp);

FIRST_NAME           JOB_ID
-------------------- --------------------
Steven               AD_PRES
Neena                AD_VP
Thomas               IT_PROG
Bruce                IT_PROG
Jennifer             AD_ASST
David                SA_REP
John                 SA_REP

7 rows selected.

SQL> |
```

4.

```
Display the employees who earn less than the least salary of DEPTNO 10.(Use
ALL operator).
```

```
SQL> select eid,first_name,sal
  2  from emp
  3  where sal > ALL (select min(sal) from emp where dept_id
=10);

       EID FIRST_NAME                  SAL
---------- -------------------- ----------
       100 Steven                    24000
       101 Neena                     17000
       104 Thomas                     6000
       105 Bruce                      4800
       106 David                      6800
       107 John                       5500

6 rows selected.
```

5.

```
Display the employees  who have the same DEPTNO and MGR of a given employee,
excluding that  employee.(Use Nested Sub Query)
```

```
SQL> select eid,first_name
  2  from emp
  3  where dept_id = (select dept_id from emp where eid=101)
  4        AND mgr = (select mgr from emp where eid=101);

       EID FIRST_NAME
---------- --------------------
       101 Neena

SQL>
```

6.

```
Write a query that displays the employee number and name of all employees who
work in a department with any employee whose name contains a R
```

```
SQL> select eid,first_name
  2  from emp
  3  where first_name IN (select first_name from emp where first_name LIKE '%r%');

       EID FIRST_NAME
---------- --------------------
       105 Bruce
       200 Jennifer
```

7.

```
The HR department needs a report that displays the ename, deptno, job of all
employees whose work in NEW YORK.(Nest Sub Query)
```

```
SQL> select first_name,dept_id,job_id
  2  from emp
  3  where dept_id = (select dept_id from departments where location_id=(select location_id from locations where city='London'));

FIRST_NAME              DEPT_ID JOB_ID
-------------------- ---------- --------------------
Steven                       90 AD_PRES
Neena                        90 AD_VP
Thomas                       60 IT_PROG
Bruce                        60 IT_PROG
Jennifer                     10 AD_ASST
David                        80 SA_REP
John                         80 SA_REP

7 rows selected.
```

8.

```
Modify the query so that the user is prompted for a LOC
```

```
SQL> select first_name,dept_id,job_id
  2  from emp
  3  where dept_id = (select dept_id from departments where location_id=(select location_id from locations where city = &city));
Enter value for city: 'Bombay'
old   3: where dept_id = (select dept_id from departments where location_id=(select location_id from locations where city = &city))
new   3: where dept_id = (select dept_id from departments where location_id=(select location_id from locations where city = 'Bombay'))

no rows selected

SQL> |
```

```
 Create a report for HR that displays the  name and salary of every employee
who reports to King.(Nest Sub Query)
```

9.

```
SQL> select first_name, sal
  2  from emp
  3  where mgr = (select eid
  4                    from emp
  5                    where last_name='KING');

FIRST_NAME                      SAL
-------------------- ----------
Neena                         17000
David                          6800

SQL> |
```

10.

```
Write a query to display all the employees working with JAMES
```

```
SQL> select first_name ,sal
  2  from emp
  3  where dept_id = (select dept_id
  4                       from emp
  5                   where last_name='JAMES');

FIRST_NAME                      SAL
-------------------- ----------
Thomas                         6000
Bruce                          4800
```

11.

```
 Display all the employees who earn less than the average salaries of their
respective departments.(Correlated Sub Queries)
```

```
SQL> select first_name,sal,dept_id
  2  from emp e
  3  where sal < ALL(select avg(sal)
  4                  from emp
  5                  where dept_id = e.dept_id);

FIRST_NAME                  SAL    DEPT_ID
-------------------- ---------- ----------
Neena                     17000         90
Bruce                      4800         60
John                       5500         80
```

12. Write a query to display the LOC and average salary of Each location.(Scalar Sub query).

```
SQL> SELECT l.city,
  2         (SELECT AVG(e.sal)
  3          FROM emp e
  4          WHERE e.dept_id IN (
  5                SELECT d.department_id
  6                FROM departments d
  7                WHERE d.location_id = l.location_id
  8          )) AS avg_salary
  9  FROM locations l;

CITY                           AVG_SALARY
------------------------------ ----------
Roma
Venice
Tokyo
Hiroshima
Southlake                            5400
South San Francisco
South Brunswick
Seattle                        15133.3333
Toronto
Whitehorse
Beijing

CITY                           AVG_SALARY
------------------------------ ----------
Bombay
Sydney
Singapore
London
Oxford                               6150
Stretford
Munich
Sao Paulo
Geneva
Bern
Utrecht

CITY                           AVG_SALARY
------------------------------ ----------
Mexico City
```

13.

```
Write a query to display the least N salaries. Use In-Line Views.
```

```
SQL> select sal from emp where ROWNUM<=3 order by sal;

       SAL
----------
      6000
     17000
     24000

SQL>
```

14.

```
Display the Last N rows from the employees table(Use Correlated Sub Queries).
```

```
SQL> select * from emp e1
  2  where 5 > (select count(*)from emp e2 where e2.eid>e1.eid)
  3  order by eid desc;

       EID FIRST_NAME           LAST_NAME                 EMAIL                     PHONE                HIRE_DATE JOB_ID                      SAL        COM
M          MGR    DEPT_ID
---------- -------------------- ------------------------- ------------------------- -------------------- --------- -------------------- ---------- ---------
- ---------- ----------
       200 Jennifer             WHALEN                    JWHALEN                   515.123.4444         17-SEP-03 AD_ASST                     4400
   101  10
       107 John                 MILLER                    JMILLER                   515.123.4571         01-JAN-08 SA_REP                      5500          .
1          106         80
       106 David                FORD                      DFORD                     515.123.4570         24-MAR-06 SA_REP                      6800          .
1          100         80
       105 Bruce                AUSTIN                    BAUSTIN                   590.423.4568         25-JUN-07 IT_PROG                     4800
   101  60
       104 Thomas               JAMES                     TJAMES                    590.423.4567         18-JUN-05 IT_PROG                     6000
   101  60
```

15.

```
Display the employees from employees table and sort only employees working in
DALLAS.(Scalar Sub Query).
```

```
SQL> SELECT eid, first_name
  2  FROM emp
  3  WHERE dept_id = (
  4      SELECT department_id
  5      FROM departments
  6      WHERE location_id = (
  7          SELECT location_id
  8          FROM locations
  9          WHERE city = 'Southlake'
 10      )
 11  )
 12  ORDER BY first_name;

       EID FIRST_NAME
---------- --------------------
       105 Bruce
       104 Thomas

SQL>
```

16.

> Display the employees who earn a salary less than avgsal of their respective department. Also display the avgsal.(Use Inline Views).

```
SQL> SELECT e.eid,
            e.first_name,
            e.dept_id,
            e.sal,
            d.avg_sal
     FROM emp e,
          (SELECT dept_id,
                  AVG(sal) AS avg_sal
           FROM emp
           GROUP BY dept_id) d
     WHERE e.dept_id = d.dept_id
       AND e.sal < d.avg_sal;

       EID FIRST_NAME              DEPT_ID        SAL    AVG_SAL
---------- -------------------- ---------- ---------- ----------
       101 Neena                        90      17000      20500
       105 Bruce                        60       4800       5400
       107 John                         80       5500       6150
```

17.

> Write a query that display the LOC of those DEPTS that have a sum of sal less than the avgsal of all the employees in DEPT table. (Use WITH Clause).

```
Connected.
SQL> with dept_sal as (
        select dept_id,
               sum(sal) as total_sal
          from emp
          group by dept_id
     ),
     avg_sal as(
        select avg(sal) as avg_salary
          from emp)
     select d.location_id
     from departments d
     join dept_sal ds
     on d.department_id = ds.dept_id
     cross join avg_sal a
     where ds.total_sal < a.avg_salary;

LOCATION_ID
-----------
       1700
```